

17 D

Minimum angle of resolution , $\theta_{\min} = \frac{\lambda}{b}$

Images seen are resolved if the angular separation between the 2 sources are larger than the minimum angle of resolution.

If the aperture size b remained unchanged, $\theta_{\min}$ will remain unchanged, and so the images will still not be resolved.

Moving the slit further from the 2 sources will decrease the angular separation of the 2 sources, and so the images will still not be resolved.

$\lambda_R > \lambda_G > \lambda_v$, using violet light will decrease the minimum angle of resolution $\theta_{\min}$, and hence allowing the angular separation between the 2 sources to be larger than $\theta_{\min}$ for the images to be resolved.